

SATHI AI & Machine Learning Architecture

A Complete, Beginner-Friendly Guide to Landslide Early Warning & Flood Management AI

What is SATHI? SATHI (Smart Automated Terrain Hazard Intelligence) is an intelligent early-warning platform designed for vulnerable regions across North-East India. It combines **satellite observation data**, **IoT ground telemetry**, and **7 specialized Machine Learning algorithms** to predict landslides before they happen and optimize emergency rescue operations when floods occur.

1. How SATHI Works (The 30,000-Foot View)

Imagine you are a disaster manager looking at an entire state like Assam, Meghalaya, or Mizoram. A landslide or flash flood happens because of three compounding factors: **weak soil/steep rock**, **heavy continuous monsoon rain**, and **topographic elevation**. Humans cannot manually monitor thousands of mountain kilometers every second. That is why SATHI uses an **Automated AI Brain** operating in three continuous steps:

- Step 1 — Sense:** Satellite radars (AlphaEarth / Sentinel) look at ground movement from space, while ground IoT sensors measure rainfall and soil moisture.
- Step 2 — Think:** The AI model takes 109 physical and environmental numbers and calculates the exact probability (0% to 100%) of a landslide occurring within the next 24 hours.
- Step 3 — Act:** The system broadcasts live alerts over WebSockets directly to the live map, sounding sirens and generating rescue priorities.

2. The Primary Landslide Early-Warning Brain (XGBoost Classifier)

The core model for landslide prediction is called **XGBoost (Extreme Gradient Boosting)**. Think of XGBoost like a committee of 100 expert scientists. The first scientist makes a prediction; the second scientist looks at where the first was mistaken and corrects it; and so on, until all 100 scientists combine their wisdom to produce a fast, highly accurate answer in under **15 milliseconds**.

The 5 Core Feature Categories Given to the AI:

Feature Category	What it Measures	Why it Matters for Landslides
1. Antecedent Rainfall	1h, 24h, 3-day, 7-day, and 14-day cumulative rainfall (mm)	62.7% Model Importance! Mountains don't collapse from 1 minute of rain; they collapse when weeks of rain saturate the soil sponge.
2. Topographic Terrain	Slope angle (°), elevation (m), aspect, and Topographic Wetness Index (TWI)	33.4% Model Importance! Steep slopes (>25°) under gravity slide much more easily than flat valleys.
3. Soil Moisture & Pore Pressure	Volumetric soil moisture at 0-7cm and 7-28cm depth layers	As water fills pore spaces between dirt particles, it creates hydraulic pressure that liquefies mud.
4. AlphaEarth Embeddings	64-dimensional feature vector from Google Earth Engine satellite scans	Captures surface vegetation health, deforestation, rock fractures, and radar displacement from space.
5. Infrastructure & History	Distance to roads, rivers, faults, and historical landslide incident count	Road excavations cut into toe-slopes of hills, making human-disturbed slopes historically more vulnerable.

How the AI Categorizes Threat (4 Risk Tiers):

LOW RISK (0 - 24) Normal conditions. Soil is dry; rainfall is manageable. Standard automated monitoring.	FLOOD WATCH (25 - 49) Rising river gauge levels or moderate slope saturation. Lowland flood advisories activated.
HIGH RISK (50 - 74) Active slope creep detected. Heavy rainfall threshold breached. Heavy earthmovers on standby.	CRITICAL DANGER (75 - 100) Imminent failure within 24h. Immediate emergency evacuation orders broadcast to civil authorities.

3. The 7 Multi-Hazard Flood & Disaster ML Modules

Beyond predicting landslides, SATHI includes a complete **Flood Disaster Management AI Suite**. When disaster strikes, authorities cannot just rely on intuition; they need mathematical optimization for triage, rescue boats, and supply distribution.

#	Disaster Module	AI Algorithm	What it Solves in Real Emergencies
1	Flood Severity Assessment	Physics & Topographic Embedding	Calculates water depth (e.g. 2.8 meters) and inundation boundaries from rainfall and river proximity.
2	Rescue Priority Identification	Random Forest Triage Classifier	Ranks neighborhoods (P1 Critical, P2 Urgent, P3 Routine) by weighing population, SOS volume, and water depth.
3	Resource Allocation Optimizer	XGBoost Regression Optimizer	Computes exact disaster relief needs: clean drinking water (litres), food rations, trauma kits, and rescue boats.
4	Area Severity Clustering	K-Means Clustering	Groups hundreds of flooded GPS points into 3-5 operational rescue clusters for NDRF command centers.

5	SOS Request Classifier	NLP Transformer (DistilBERT)	Reads frantic citizen SMS/social posts, extracts urgency, and categorizes into Medical, Evacuation, or Food needs.
6	Damage Assessment	Structural Hydrodynamic Model	Estimates collapse hazard for bridges, concrete buildings, and roads based on water flow velocity.
7	Safe Route Recommendation	Dijkstra / A* Graph Engine	Computes rescue paths between origin and shelters while dynamically steering away from submerged danger zones.

4. The End-to-End Real-Time Pipeline

Here is the exact journey of a single sensor reading through the software stack:

1. Ingestion	An IoT hill station transmits telemetry: <i>Temperature: 26°C, Humidity: 85%, Soil: 48%, Rain: 45mm.</i>
2. Feature Assembly	The FastAPI backend combines this live reading with historical rainfall windows (3d, 7d, 14d) and GIS slope maps.
3. Model Inference	The XGBoost model evaluates the 109-element feature vector and outputs a Risk Score: 88 (CRITICAL) in <15ms.
4. Database Save	The prediction record and timestamp are persisted in MySQL (or SQLite offline store) for historical trend auditing.
5. WebSocket Broadcast	The native WebSocket server (<code>`/ws/risk`</code>) pushes the live JSON event to all connected React dashboards and live maps.
6. Map UI Update	The user's web browser pulses the affected mountain polygon red, sounds alerts, and lists actionable evacuation steps.

5. Summary Glossary for Beginners

- **Machine Learning (ML):** Teaching a computer to recognize patterns from thousands of past landslide events instead of hardcoding static if-else rules.
- **Inference:** The moment the trained AI looks at new live sensor data and calculates a risk prediction.
- **Embeddings (AlphaEarth):** Translating high-resolution satellite imagery into a compact list of 64 meaningful numbers representing ground health.
- **Antecedent Rainfall:** Rain that fell days or weeks ago that remains trapped underground, saturating rock layers.
- **WebSocket:** A two-way open internet pipe allowing the server to push live data to your screen instantly without having to refresh the page.